Memory Hierarchy

- Memory Hierarchy is one of the most required things in Computer Memory as it helps in optimizing the memory available in the computer.

There are multiple levels present in the memory, each one having a different size, different cost, etc.

- Some types of memory like cache, and main memory are faster as compared to other types of memory but they are having a little less size and are also costly whereas some memory has a little higher storage value, but they are a little slower.
- Accessing of data is not similar in all types of memory, some have faster access whereas some have slower access.

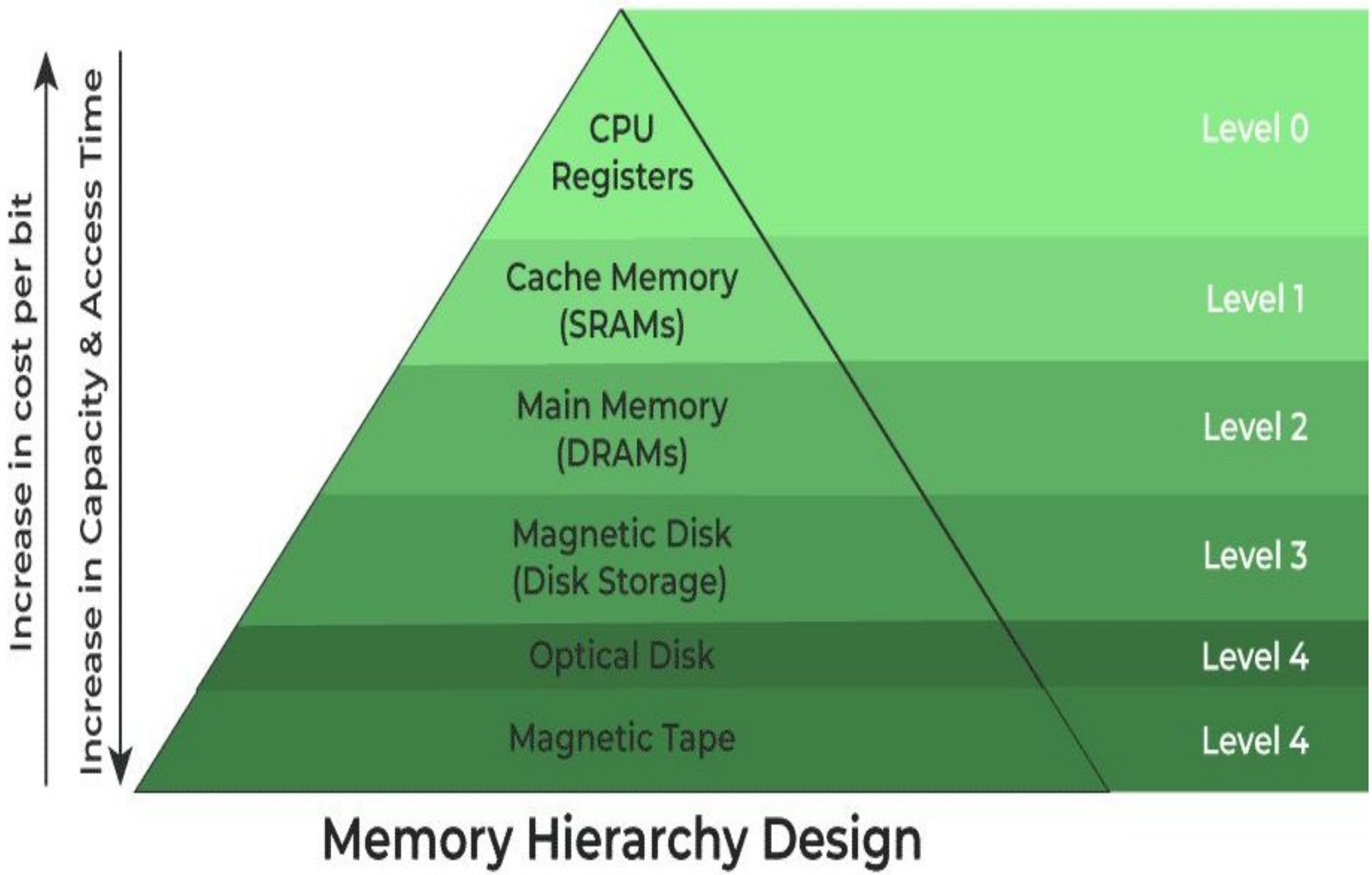
Types of memory hierarchy

- **External Memory or Secondary Memory:** Comprising of Magnetic Disk, Optical Disk, and Magnetic Tape i.e. peripheral storage devices which are accessible by the processor via an I/O Module.
- **Internal Memory or Primary Memory:** Comprising of Main Memory, Cache Memory & CPU registers. This is directly accessible by the processor.
- The memory technology and storage organization at each level are characterized by five parameters:

i. Access time.
ii. Memory size.
iii. Cost per byte.
iv. Transfer bandwidth.
v. Unit of transfer.

❖ **The unit of transfer in memory hierarchy is the number of data lines that move into and out of a memory module. It can also refer to the number of bits that can be read or written to memory at a time.**

Four level memory hierarchy Design



Memory Hierarchy Design

- **1. Registers**
- Registers are small, high-speed memory units located in the CPU. They are used to store the most frequently used data and instructions. Registers have the fastest access time and the smallest storage capacity, typically ranging from 16 to 64 bits.
- **2. Cache Memory**
- Cache memory is a small, fast memory unit located close to the CPU. It stores frequently used data and instructions that have been recently accessed from the main memory. Cache memory is designed to minimize the time it takes to access data by providing the CPU with quick access to frequently used data.
- **3. Main Memory**
- Main memory also known as RAM (Random Access Memory), is the primary memory of a computer system. It has a larger storage capacity than cache memory, but it is slower. Main memory is used to store data and instructions that are currently in use by the CPU.

Types of Main Memory

- **Static RAM:** it stores the binary information in flip flops and information remains valid until power is supplied. It has a faster access time and is used in implementing cache memory.

4. Secondary Storage

- Secondary storage, such as hard disk drives (HDD) and solid-state drives (SSD) , is a non-volatile memory unit that has a larger storage capacity than main memory. It is used to store data and instructions that are not currently in use by the CPU.
- Secondary storage has the slowest access time and is typically the least expensive type of memory in the memory hierarchy.
- **5. Magnetic Disk**
- Magnetic Disks are simply circular plates that are fabricated with either a metal or a plastic or a magnetized material. The Magnetic disks work at a high speed inside the computer and these are frequently used.
- **6. Magnetic Tape**
- Magnetic Tape is simply a magnetic recording device that is covered with a plastic film. It is generally used for the backup of data. In the case of a magnetic tape, the access time for a computer is a little slower and therefore, it requires some amount of time for accessing the strip.

Advantages of Memory Hierarchy

- It helps in removing some destruction, and managing the memory in a better way.
- It helps in spreading the data all over the computer system.
- It saves the consumer's price and time.